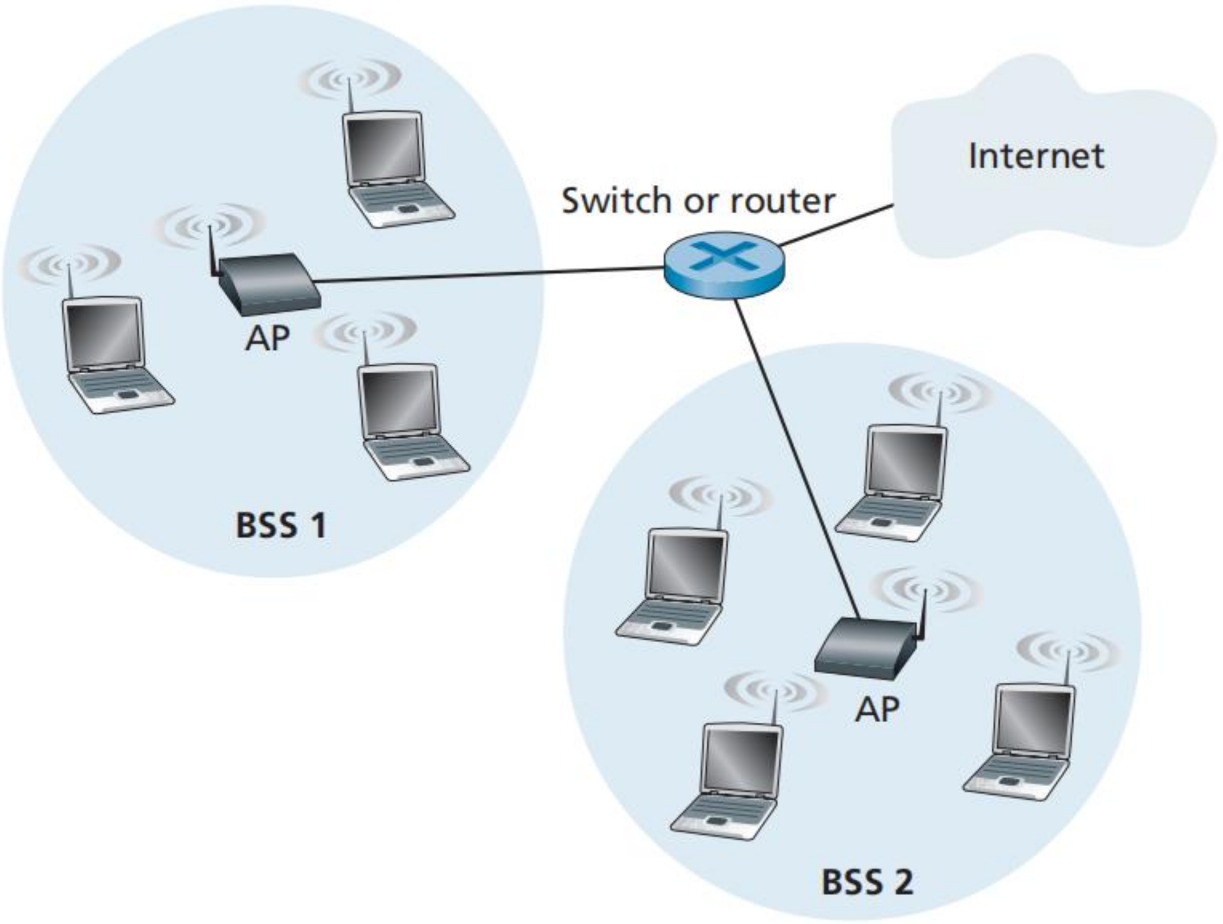
Introduction to Internet of Things

Lecture 17&18

802.11 Wireless LAN Architecture



- Basic Service Set (BSS)
- Wireless Stations
- Base Station = Access Point

IEEE 802.11 Infrastructure WLAN

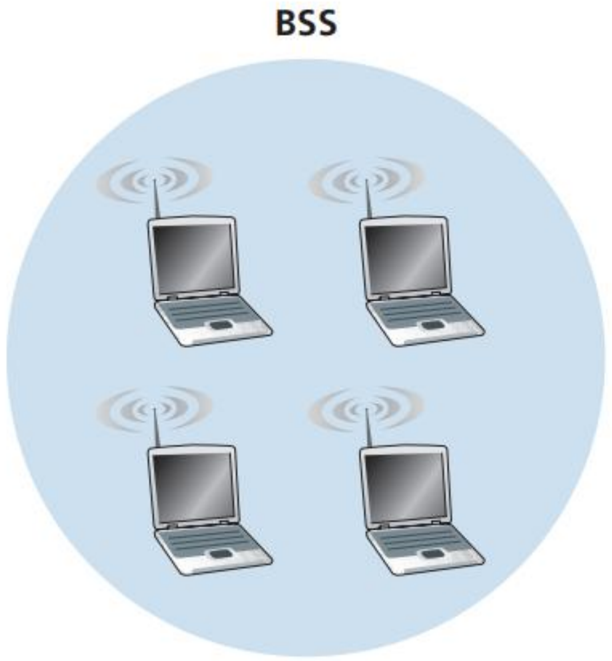
- 6 byte MAC address is stored in the firmware of the station's adapter.
- Each AP has also a MAC address for its wireless interface.
- APs have the wired connection with router.

IEEE 802.11 Ad-hoc networks

- A network with no central control and with no connections to the outside world.

Examples:

1. Military or disaster communication networks: Soldiers' radios form a mobile ad-hoc network (MANET).
2. People with laptops get together (e.g., in a conference room, a train, or a car) and want to exchange data in the absence of a centralized AP.
3. IoT sensors in remote areas: Sensors communicate directly with nearby sensors.



Channels and Association in WLAN

- Network Administrator assigns SSID (Service Set Identifier) and channel no. to the AP.
- 802.11 operates in the frequency range of 2.4 GHz to 2.4835 GHz.
- Within this 85 MHz band, 802.11 defines 11 partially overlapping channels.
- Any two channels are non-overlapping if and only if they are separated by four or more channels.
- The set of channels 1, 6, and 11 is the only set of three non-overlapping channels.
- Associating means the wireless device creates a virtual wire between itself and the AP.

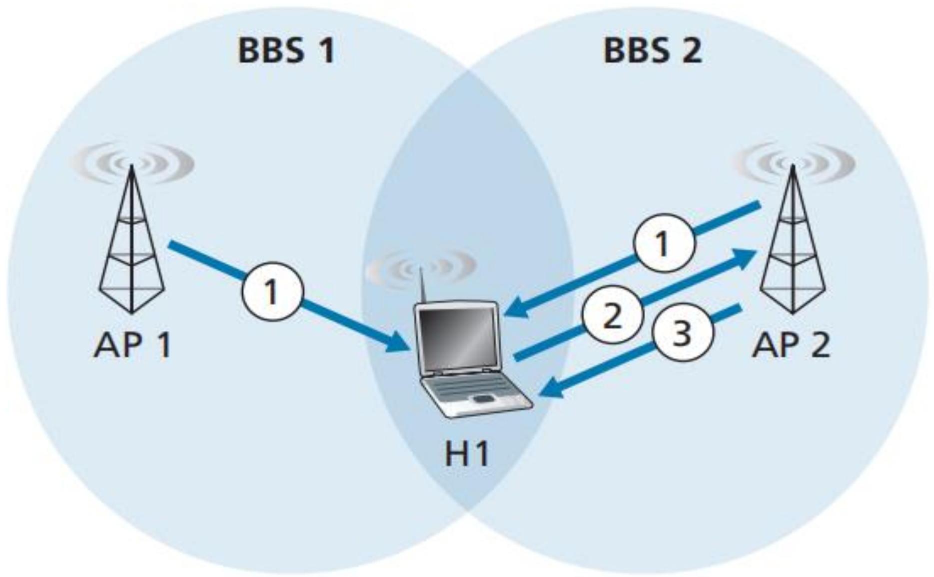
WiFi Jungle

- A WiFi jungle is any physical location where a wireless station receives a sufficiently strong signal from two or more APs.
- Each of these APs would likely be located in a different IP subnet and would have been independently assigned a channel.

  How does your wireless device associate with a particular AP?

- The 802.11 standard requires that an AP periodically send beacon frames, each of which includes the AP's SSID and MAC address.
- Your wireless device scans all 11 channels in 2.4 GHz band.
- It listens for beacon frames from all APs it can detect.
- If multiple APs transmit on the same channel, your device will see all of them.

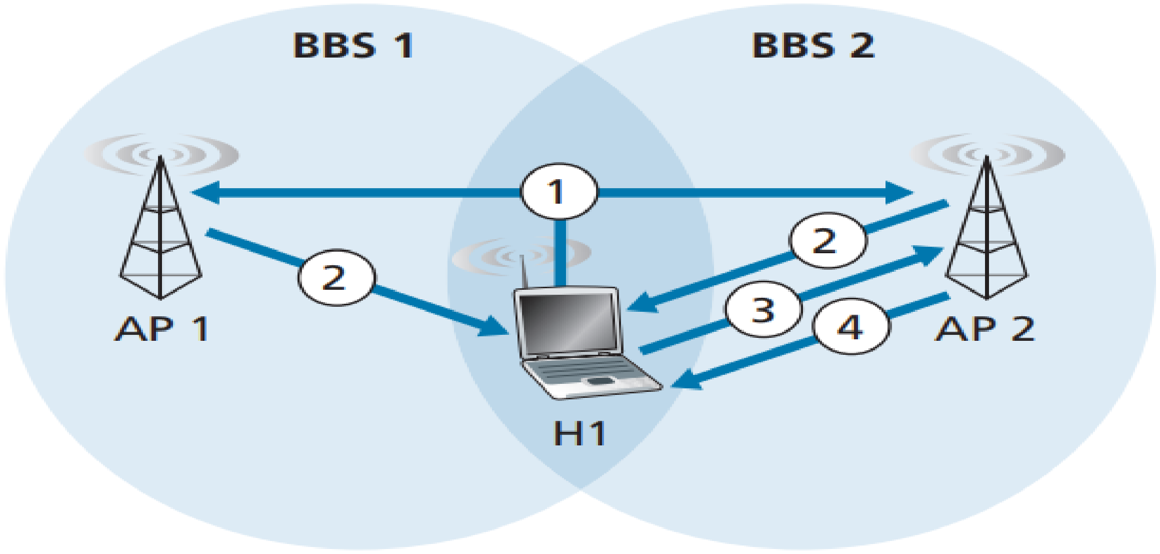
- Typically, the device chooses the AP whose beacon frame is received with the highest signal strength.
- The process of scanning channels and listening for beacon frames is known as **passive scanning.**



a. Passive scanning

1. Beacon frames sent from APs
2. Association Request frame sent: H1 to selected AP
3. Association Response frame sent: Selected AP to H1

- A wireless device can also perform **active scanning**, by broadcasting a probe frame that will be received by all APs within the wireless device’s range,



a. Active scanning

1. Probe Request frame broadcast from H1
2. Probes Response frame sent from APs
3. Association Request frame sent: H1 to selected AP
4. Association Response frame sent: Selected AP to H1

802.11 MAC Protocol

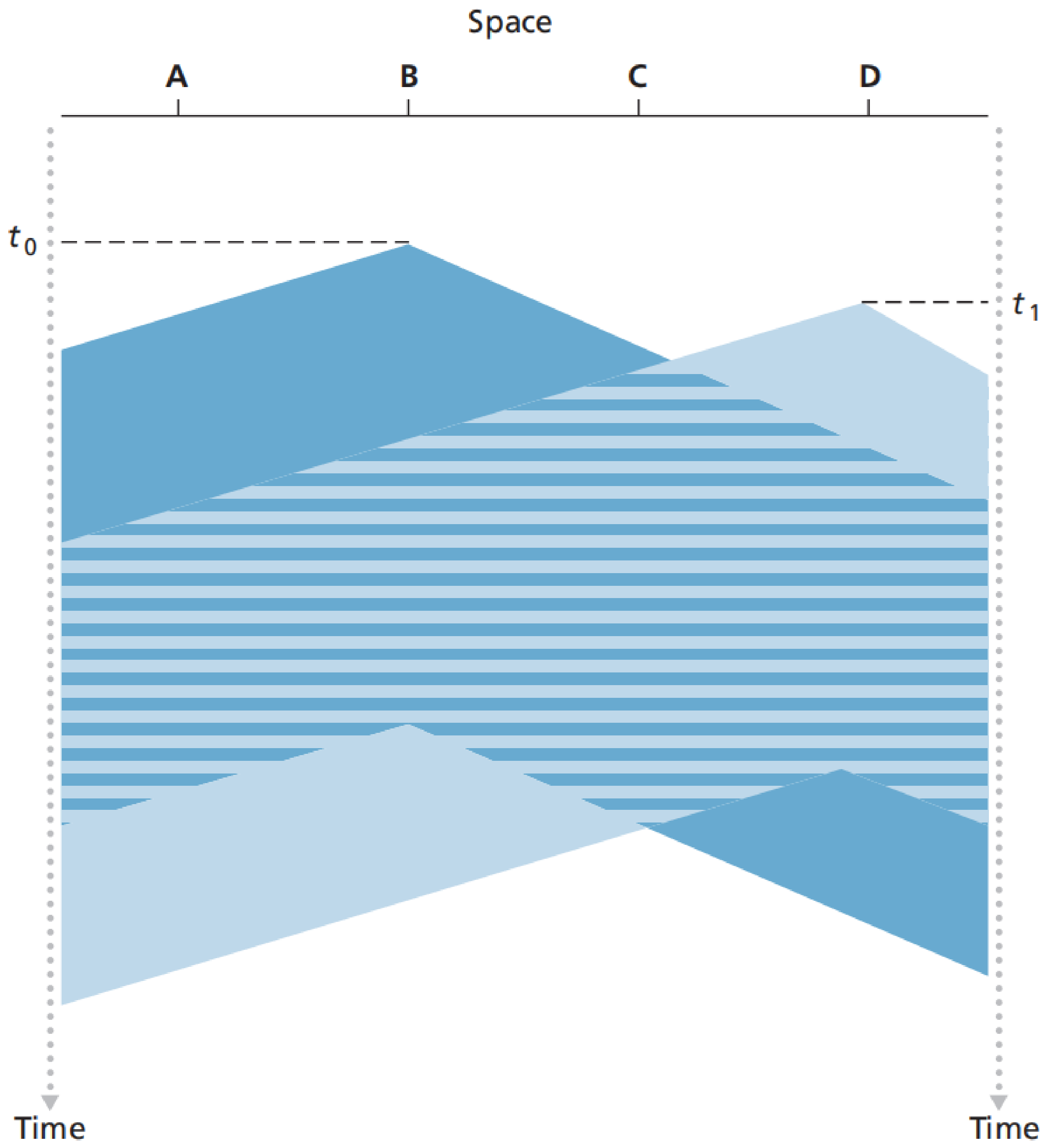
- There are three classes of multiple access protocols:

1. Channel partitioning
2. Random access
3. Taking turns

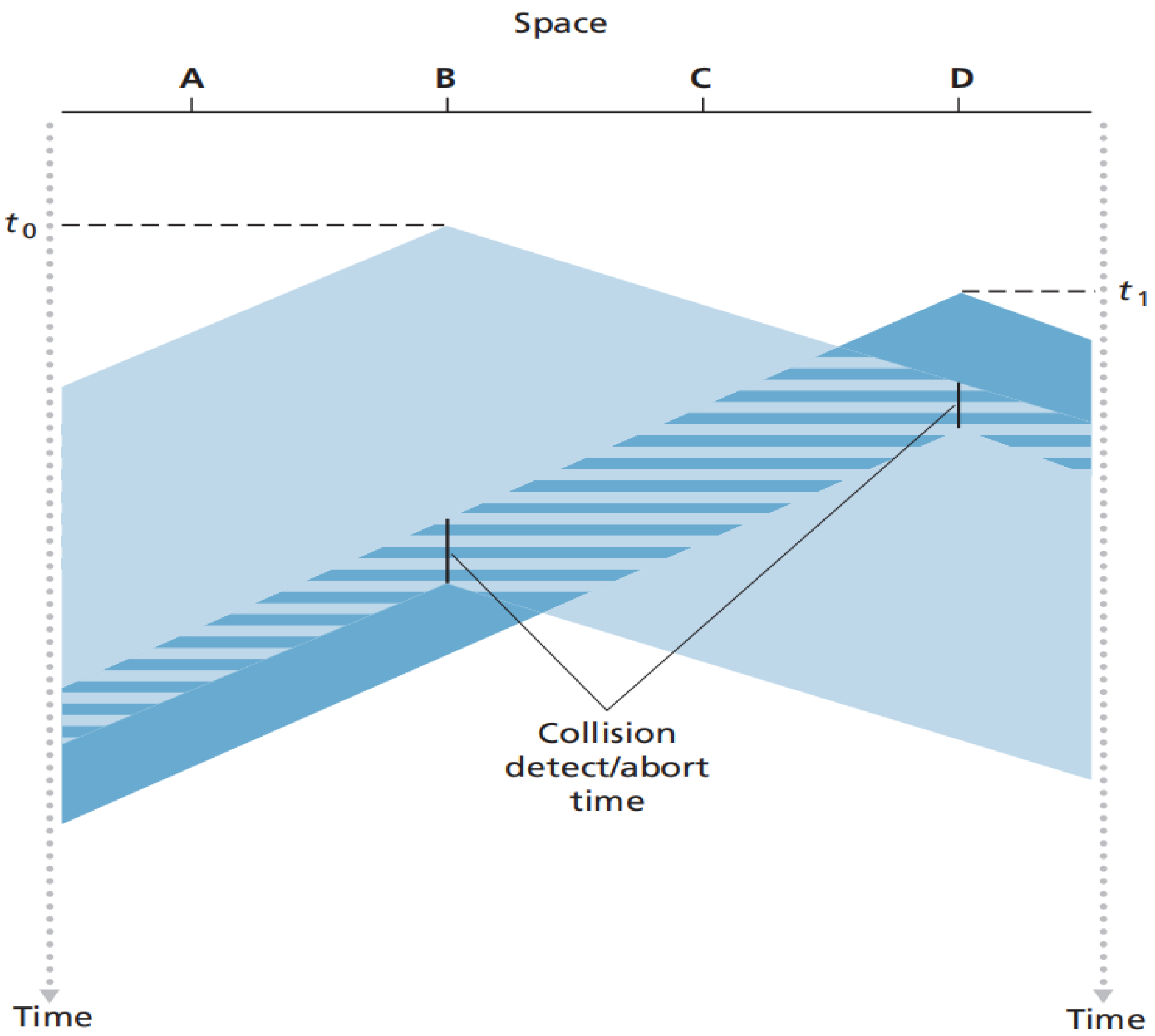
- Random access protocol is used in 802.11 that is CSMA/CA, CSMA with collision avoidance.
- 802.11 uses collision-avoidance techniques and because of the relatively high bit error rates of wireless channels, it uses a link-layer acknowledgment/retransmission (ARQ) scheme.

CSMA

- CSMA: Carrier Sensing
- CSMA/CD: Collision Detection



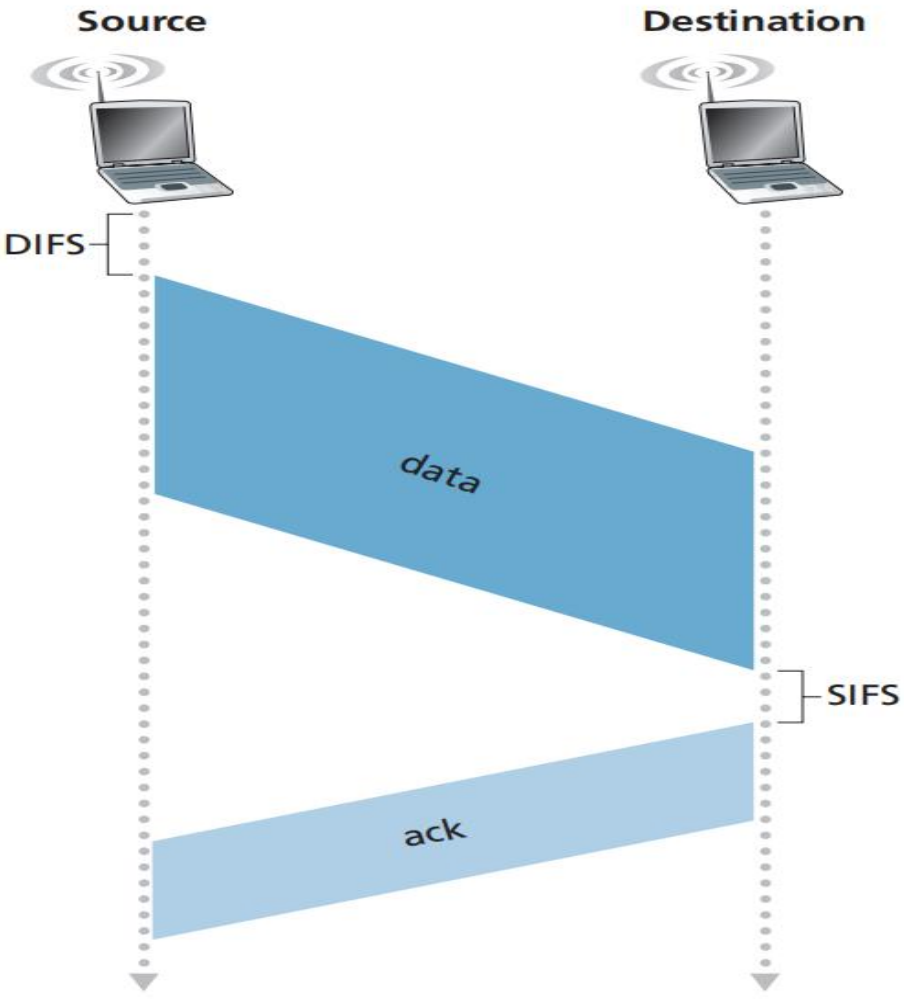
CSMA/CD



Binary Exponential Backoff

- When transmitting a frame that has already experienced n collisions, a node chooses the value of K at random from {0,1,2, ... 2^n-1}.
- Thus, the more collisions experienced by a frame, the larger the interval from which K is chosen.
- For Ethernet, the actual amount of time a node waits is K * 512 bit times.

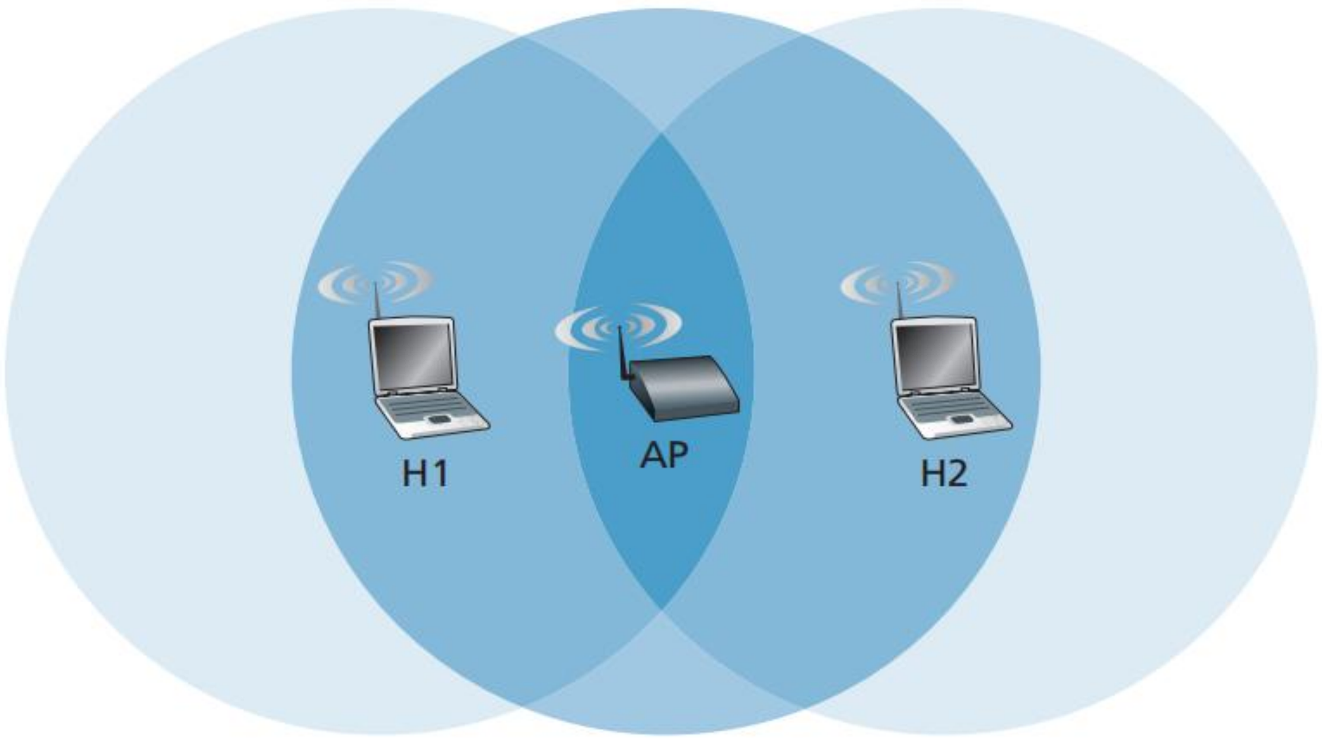
Link Layer Acknowledgment Scheme



CSMA/CA protocol

1. If initially the station senses the channel idle, it transmits its frame after a short period of time known as the Distributed Inter-frame Space (DIFS).
2. Otherwise, the station chooses a random backoff value using binary exponential backoff and counts down this value after DIFS when the channel is sensed idle. While the channel is sensed busy, the counter value remains frozen.
3. When the counter reaches zero, the station transmits the entire frame and then waits for an acknowledgment.
4. If the station has another frame to send, it begins the CSMA/CA protocol at step 2. If the acknowledgment isn't received, the transmitting station reenters the backoff phase in step 2, with the random value chosen from a larger interval.

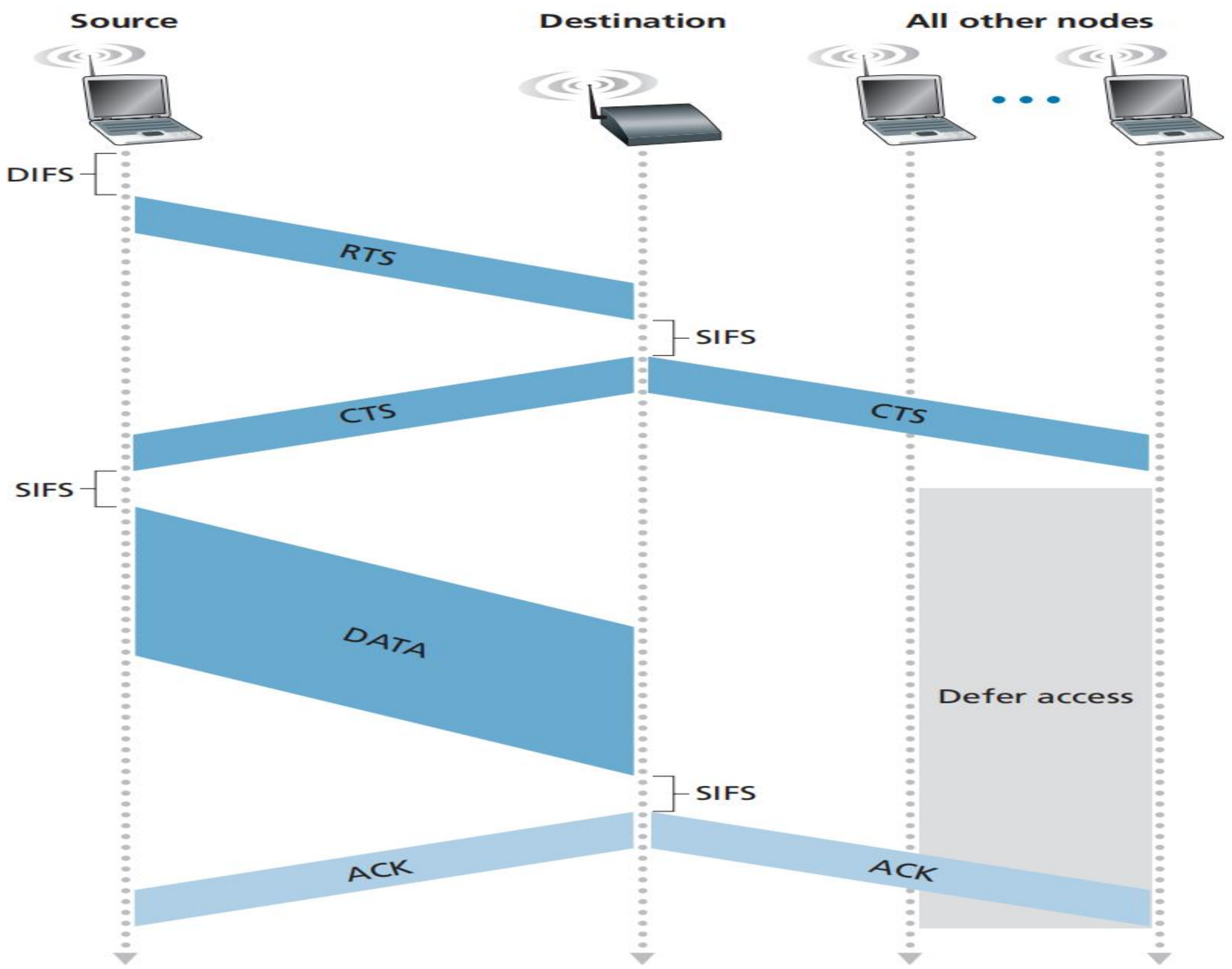
Hidden Terminal Problem



Solution

1. Request to Send (RTS) control frame
2. Clear to Send (CTS) control frame

- When a sender wants to send a DATA frame, it can first send an RTS frame to the AP, indicating the total time required to transmit the DATA frame and the acknowledgment (ACK) frame.
- When the AP receives the RTS frame, it responds by broadcasting a CTS frame.


Source
Destination
All other nodes
DIFS
RTS
SIFS
CTS
CTS
SIFS
DATA
Defer access
SIFS
ACK
ACK

802.11 Frame

Frame (numbers indicate field length in bytes):

2	2	6	6	6	2	6	0-2312	4
Frame control	Duration	Address 1	Address 2	Address 3	Seq control	Address 4	Payload	CRC

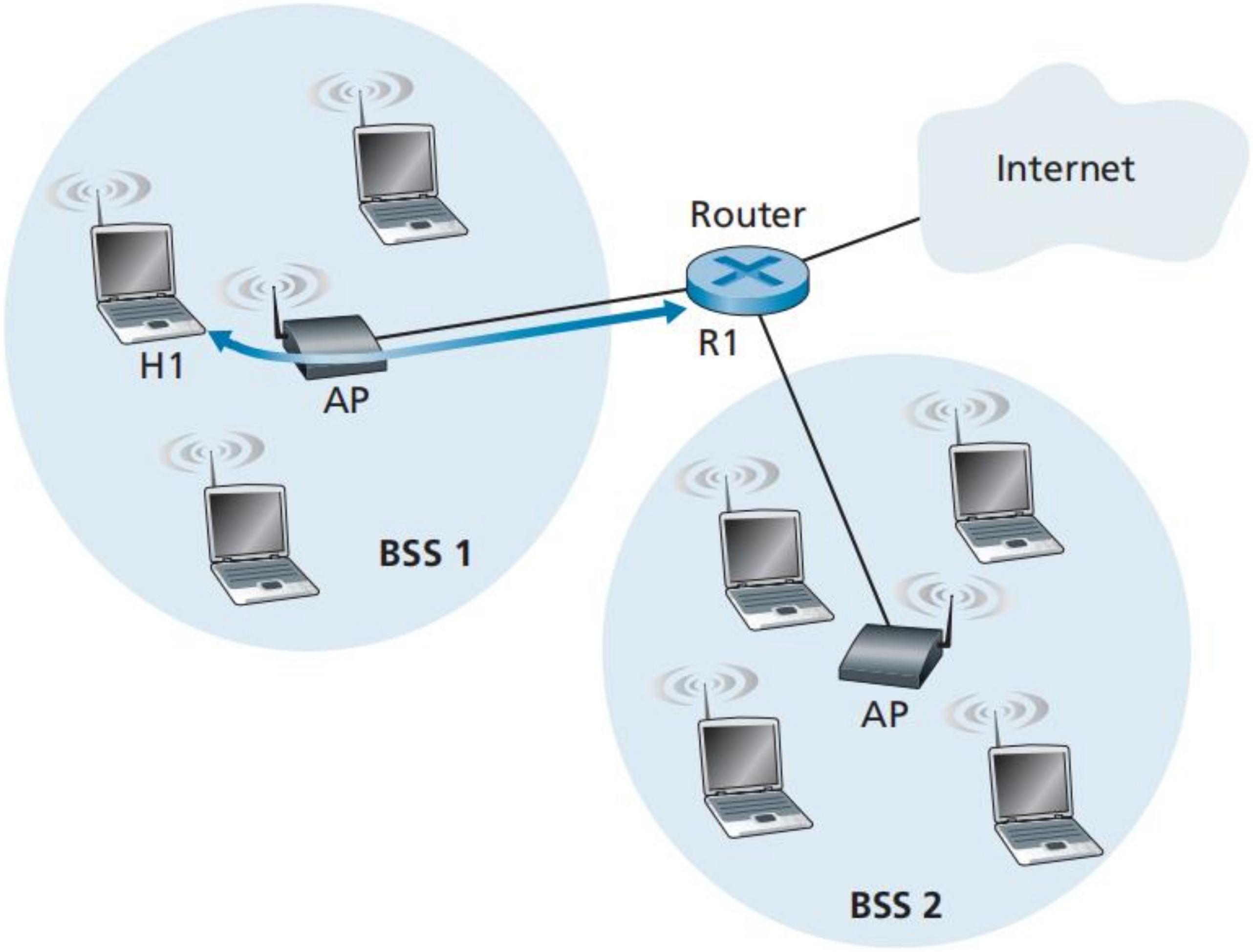
Frame control field expanded (numbers indicate field length in bits):

2	2	4	1	1	1	1	1	1	1	1
Protocol version	Type	Subtype	To AP	From AP	More frag	Retry	Power mgt	More data	WEP	Rsvd

802.11 Frame

Why 4 address fields?

- The fourth address field is used when APs forward frames to each other in ad hoc mode.
- The second address field holds the MAC address of transmitting node.
- The first address field holds the MAC address of receiving node.
- BSS is part of a subnet connected to any router interface. Address 3 contains the MAC address of this router interface.


Internet
Router
R1
H1
AP
BSS 1
AP
BSS 2

- The router, which knows the IP address of H1 uses ARP to determine the MAC address of H1.
- After obtaining H1's MAC address, router interface R1 encapsulates the datagram within an Ethernet frame.
- When the Ethernet frame arrives at the AP, the AP converts the 802.3 Ethernet frame to an 802.11 frame before transmitting the frame into the wireless channel.
- For address 3, the AP inserts the MAC address of R1. In this manner, H1 can determine (from address 3) the MAC address of the router interface that sent the datagram into the subnet.

Address 3 plays a crucial role for internetworking the BSS with a wired LAN.

What happens when H1 responds to R1?